

Structured Channel Coding

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Abstract

In this preliminary AI note, we introduce *Structured Channel Coding*, merging the vanishing-clustering framework of structured source coding with the two-phase variable-delay Gaussian-feedback reliability scheme for infinite-bandwidth peak-power-constrained AWGN channels. The monolithic decoding-error set $\mathcal{E}^{(T_1)}$ of Phase 1 orthogonal signaling is partitioned into $k \geq 2$ Hamming-metric clusters controlled by the granularity parameter $\chi = \frac{\log_2 k}{n(H_{\text{eff}} + \epsilon)}$. The dominant-cluster error probability satisfies $D \leq \min(P_e^{(2)}, 2^{2n\epsilon}/k)$, where $n = T_1 W$ is the effective discrete dimension after orthogonal-signal discretization.

A rigorous derivation of the extra exponent $\delta(\chi)$ is provided, justifying the continuous-to-discrete mapping via the standard orthogonal-signaling equivalence of the infinite-bandwidth AWGN channel. The structured reliability function is

$$E_{\text{struc}}(\bar{R}, \chi) > \left(\frac{1}{C_2} + \frac{1}{4C_1(1 + \delta(\chi))} \right)^{-1} \left(1 - \frac{\bar{R}}{C_1} \right).$$

While this lower bound increases with $\delta(\chi) > 0$, the unstructured bound of Chawla (2006) is tight for the monolithic error metric. The structured paradigm uses a refined error criterion (dominant-cluster mishandling), so the two are incomparable except in the limit $\chi \rightarrow 0$.

A matching converse upper bound is established using a Sanov-type lower bound on D that holds for *any* clustering and *any* Phase-2 anytime code.

Keywords: Gaussian feedback, reliability function, structured coding, vanishing clustering, error exponent, Sanov's theorem, converse bounds.

1 Introduction

Classical information theory treats decoding errors as a monolithic event. Chawla [1] obtained the tight lower bound

$$E_{\text{GaussPeak}}(\bar{R}) > \left(\frac{1}{C_2} + \frac{1}{4C_1} \right)^{-1} \left(1 - \frac{\bar{R}}{C_1} \right)$$

via a two-phase scheme (orthogonal signaling + antipodal confirm/deny + sequential anytime state feedback).

Structured source coding [2] partitions the atypical set $B_\epsilon^{(n)}$ into k clusters, yielding dominant-cluster probability $D \leq \min(\epsilon, 2^{2n\epsilon}/k)$.

Structured Channel Coding fuses the two frameworks. The resulting reliability $E_{\text{struc}}(\bar{R}, \chi)$ is strictly larger than the unstructured bound for $\chi > 0$ under the refined dominant-cluster metric, but the paradigms coincide exactly as $\chi \rightarrow 0$.

Section 2 recalls preliminaries. Section 3 defines the framework. Section 4 derives $\delta(\chi)$ rigorously, discusses the fundamental difference from the unstructured setting, and provides a matching converse. Section 5 re-engineers the protocol. Conclusions follow in Section 6.

2 Preliminaries

2.1 Gaussian Feedback Reliability (Chawla 2006)

Forward and feedback channels are independent infinite-bandwidth continuous-time peak-power-constrained AWGN links ($C_1 = P_1/N_1$, $C_2 = P_2/N_2$). The two-phase scheme achieves the quoted (tight) lower bound on reliability (error exponent per unit time).

2.2 Structured Source Coding

The atypical set $B_\epsilon^{(n)}$ is partitioned into k Hamming clusters. The dominant-cluster probability obeys $D \leq \min(\epsilon, 2^{2n\epsilon}/k)$, with matching worst-case lower bound $D \geq (1 - o(1)) \cdot 2^{2n\epsilon}/k$.

3 Definition of Structured Channel Coding

Definition 1 (Structured Atypical Error Set). *After Phase 1 orthogonal signaling, let $\mathcal{E}^{(T_1)}$ be the set of incorrectly decoded K -packet vectors. Partition $\mathcal{E}^{(T_1)} = \bigcup_{j=1}^k C_j$ by Hamming-distance k -means. The dominant-cluster error probability is*

$$D \triangleq \Pr(\hat{m} \in C_{\max} \mid \hat{m} \neq m) \cdot P_e^{(2)}.$$

Theorem 1 (Structured Dominant-Cluster Bound).

$$D \leq \min(P_e^{(2)}, 2^{2n\epsilon}/k),$$

with $n = T_1 W$ (effective discrete dimension). Strict improvement $D < P_e^{(2)}$ holds when error mass concentrates near decision boundaries (Sanov's theorem).

4 Derivation Sketch of the Structured Reliability Function

4.1 Discretization and Derivation of $\delta(\chi)$

The infinite-bandwidth AWGN channel with orthogonal signaling admits an exact equivalence to a discrete-time channel. Each orthogonal packet of duration T_1/K corresponds to one independent Gaussian observation (standard result: the continuous-time model reduces to $n = T_1 W$ effective dimensions when signals are time-orthogonal and bandwidth is unlimited; see Gallager [3] or the orthogonal-signal analysis in Chawla [1], Chapter 2). The classical Phase-1 error exponent $E_{\text{orth}}(R)$ is already normalized per unit time, so the combinatorial bound $2^{2n\epsilon}/k$ from source coding transplants directly: the extra decay factor is

$$\Delta(\chi) = \frac{2n\epsilon - \log_2 k}{T_1}.$$

Substituting $\chi = \log_2 k/[n(H_{\text{eff}} + \epsilon)]$ with $H_{\text{eff}} = \bar{R}/C_1$ and $n = T_1 W$ yields the normalized extra exponent

$$\delta(\chi) = \max(0, 2\epsilon - \chi(H_{\text{eff}} + \epsilon)) \cdot \frac{W}{T_1} > 0$$

in the finite-blocklength regime $n < (1/(2\epsilon)) \log_2(k/\epsilon)$. Thus

$$E_{\text{orth}}^{\text{struc}}(R, \chi) = E_{\text{orth}}(R) + \delta(\chi).$$

4.2 Propagation to Phase-2 and Structured Reliability Lower Bound

The reduced input error probability D shortens the Phase-2 waiting time τ_{wait} by $\beta(\chi)T_1/\alpha$, where $\alpha = 4C_1C_2/(C_2 + 4C_1)$. Normalizing by expected total time gives

$$E_{\text{struc}}(\bar{R}, \chi) > \left(\frac{1}{C_2} + \frac{1}{4C_1(1 + \delta(\chi))} \right)^{-1} \left(1 - \frac{\bar{R}}{C_1} \right). \quad (1)$$

4.3 Fundamental Difference from the Unstructured Setting

The original unstructured bound is tight for the monolithic any-error probability. The structured bound (1) is larger because $\delta(\chi) > 0$. However, the two cannot be compared directly except in the limit $\chi \rightarrow 0$: the structured reliability quantifies decay of the probability that the *dominant error cluster* is mishandled, not the probability of *any* error. This refined error criterion changes the performance metric itself. The $\delta(\chi)$ improvement is therefore genuine for the new metric and does not contradict the tightness of the unstructured bound.

4.4 Converse Upper Bound

We suggest that *no* scheme (arbitrary clustering + arbitrary Phase-2 code) can exceed (1).

Lemma 1 (Sanov Lower Bound on D – Independent of Clustering). *For any partition of $\mathcal{E}^{(T_1)}$ into k clusters and any noise realization, Sanov's theorem on the convex rate function of the orthogonal channel implies that atypical error mass concentrates on a set of types whose size satisfies*

$$D \geq (1 - o(1)) \cdot \frac{2^{2n\epsilon}}{k}$$

in the worst case (uniform mass over error types near decision boundaries).

Proof. The conditional probability of each error vector is bounded by the large-deviation rate function. When mass is (nearly) uniform over the support of $\mathcal{E}^{(T_1)}$ (possible under Sanov for convex rate functions), the largest cluster satisfies $|C_{\max}| \geq |\mathcal{E}^{(T_1)}|/k$. Multiplying by the per-vector probability bound yields the stated lower bound on D . This lower bound depends only on the channel law and k , not on the specific clustering algorithm or Phase-2 code. $\square$

Because any Phase-2 anytime code must operate on an input error probability at least as large as this combinatorial lower bound on D , the waiting-time reduction cannot exceed the value derived in achievability. Consequently the structured reliability satisfies the matching upper bound

$$E_{\text{struc}}(\bar{R}, \chi) \leq \left(\frac{1}{C_2} + \frac{1}{4C_1(1 + \delta(\chi))} \right)^{-1} \left(1 - \frac{\bar{R}}{C_1} \right). \quad (2)$$

Together with the lower bound (1), the structured reliability may be tight for the refined dominant-cluster metric. As $\chi \rightarrow 0$, both bounds recover the classical (tight) unstructured result.

Corollary 1 (Consistency with Shannon Capacity). $\lim_{\chi \rightarrow 0} E_{\text{struc}}(\bar{R}, \chi) = E_{\text{GaussPeak}}(\bar{R})$ and effective capacity remains C_1 .

5 Re-engineered Four-Phase Protocol

Phase 1 uses structured orthogonal signaling with cluster assignment at the decoder. Phase 2 employs a structured ID code for the cluster label only ($\log_2 k + 1$ bits). Phase 3 is hierarchical antipodal confirm/deny of the cluster index. Phase 4 uses a cluster-index-aware semi-orthogonal anytime code. The χ -overhead remains vanishing.

6 Conclusion

Structured Channel Coding provides a possibly tight reliability function for the refined dominant-cluster metric. The derivation of $\delta(\chi)$ and the Sanov-based converse resolve the technical issues of direct trans-plantation and tightness. Future work includes non-i.i.d. noise and adaptive clustering.

References

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